NEWTON'S LAWS

Newton's Three Laws

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PHYSICS

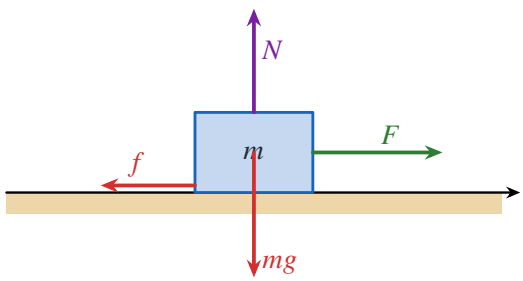
1 First Law: Inertia

$$\sum \vec{F} = 0 \iff \vec{v} = \text{constant}$$

A body keeps its state of rest or uniform motion unless a net force acts. Holds only in inertial frames.

2 Second Law

$$\vec{F} = \frac{d\vec{p}}{dt} = m\vec{a}$$



Draw every force on one body: here $F - f = ma$ and $N = mg$.

3 Third Law

$$\vec{F}_{AB} = -\vec{F}_{BA}$$

Action and reaction act on different bodies, so they never cancel each other.

4 Impulse

$$J = \int F\,dt = \Delta p$$

Same change in momentum over a longer time means a smaller force: why we bend our knees.

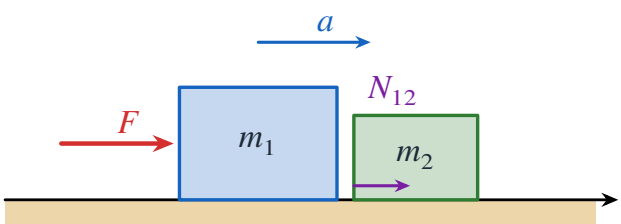
NEWTON'S LAWS

Connected Bodies • Lift

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5 Blocks in Contact

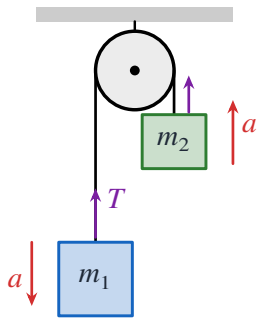
$$a = \frac{F}{m_1 + m_2}, \qquad N_{12} = m_2 a$$



$F = 30$ N on 4 kg + 2 kg: $a = 5$ m/s^2, $N_{12} = 10$ N.

6 Atwood Machine

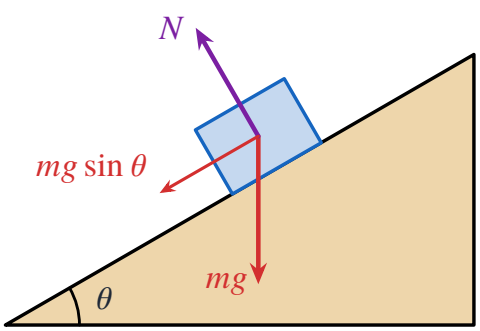
$$a = \frac{(m_1 - m_2)g}{m_1 + m_2}, \qquad T = \frac{2m_1 m_2 g}{m_1 + m_2}$$



5 kg and 3 kg: $a = 2.5$ m/s^2, $T = 37.5$ N.

7 Smooth Incline

$$a = g \sin\theta, \qquad N = mg \cos\theta$$



Resolve along and across the slope, not horizontal and vertical.

8 Apparent Weight in a Lift

$$N = m(g + a)$$

a up: heavier; a down: lighter; free fall ($a = -g$): $N = 0$. 60 kg reads 72 / 60 / 48 kg.

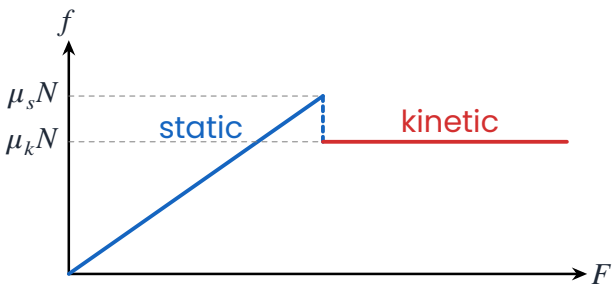
NEWTON'S LAWS

Friction

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9 Static and Kinetic

$$f_s \le \mu_s N, \qquad f_k = \mu_k N$$



Static friction adjusts up to $\mu_s N$; then the block slips and $\mu_k N$ ($< \mu_s N$) acts.

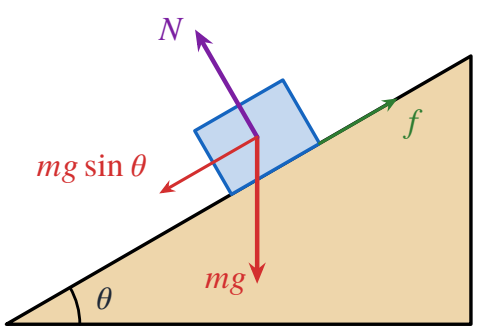
10 Angle of Repose

$$\tan\theta_r = \mu_s$$

Tilt the plane slowly: the block starts to slide when the slope reaches this angle.

11 Rough Incline

$$a = g(\sin\theta - \mu_k \cos\theta)$$



Friction points up the slope while the block slides down.

12 Least Pulling Force

$$F_{\min} = \frac{\mu mg}{\sqrt{1+\mu^2}} \quad \text{at} \quad \tan\phi = \mu$$

Pulling a little upwards lowers N, and so the friction.

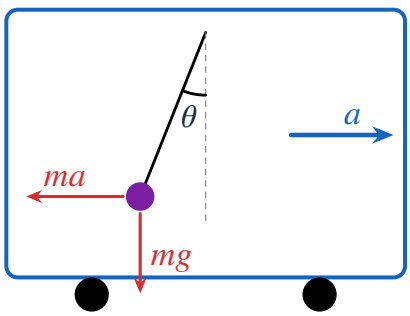
NEWTON'S LAWS

Pseudo Force • Circular Motion

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13 Pseudo Force

$$\vec{F}_{\text{pseudo}} = -m\vec{a}_{\text{frame}}$$



In a car speeding up at a, a hanging bob tilts back by $\tan\theta = a/g$.

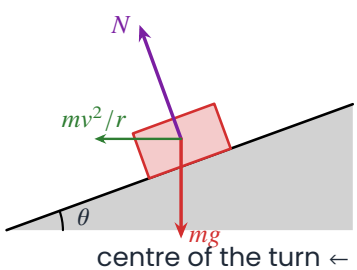
14 Circular Motion

$$F_c = \frac{mv^2}{r} = m\omega^2 r$$

Not a new force: tension, friction, gravity or N supplies it.

15 Banked Road

$$\tan\theta = \frac{v^2}{rg}$$



centre of the turn ←

No friction needed at this speed: $N\sin\theta$ gives the centripetal force.

16 Flat Road Limit

$$v_{\max} = \sqrt{\mu rg}$$

On a flat turn only friction keeps the car on the circle.

NEWTON'S LAWS

Quick Revision • Traps

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17 Every Result at a Glance

Situation	Result
Blocks in contact	$a = F/(m_1 + m_2)$, $N_{12} = m_2 a$
Atwood machine	$a = (m_1 - m_2)g/(m_1 + m_2)$
Lift	$N = m(g + a)$
Incline, smooth / rough	$g \sin\theta$ / $g(\sin\theta - \mu\cos\theta)$
Angle of repose	$\tan\theta = \mu_s$
Banked road	$\tan\theta = v^2/rg$

18 Common Traps

- Action-reaction pairs act on different bodies.
- Static friction is not always $\mu_s N$: only at the limit.
- Pseudo force only in an accelerating frame.
- Centripetal force is not an extra force on the FBD.

19 Quick Numerical

Atwood: 5 kg and 3 kg, $g = 10$. Find a and T.

$$a = 2.5\ \text{m/s}^2, \quad T = 37.5\ \text{N}$$